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R. Manoj Kumar, Sandan Kumar Sharma, B.V. Manoj Kumar, Debrupa Lahiri

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Effects of Carbon Nanotube Aspect Ratio on Strengthening and Tribological Behaviour of Ultra High Molecular Weight Polyethylene Composite

R Manoj Kumar^{1,2}, Sandan Kumar Sharma², B.V. Manoj Kumar², Debrupa Lahiri^{1,2} & *

¹Biomaterials and Multiscale Mechanics Lab, ²Department of Metallurgical and Materials Engineering

Indian Institute of Technology Roorkee, Roorkee – 247667, India

* Correspondence Author. Tel: 91-1332-28-5137. E-mail: debrupa.lahiri@gmail.com (Debrupa Lahiri)

Abstract

Carbon-based nanomaterials are great choice as reinforcement to ultra-high molecular-weight polyethylene (UHMWPE), with potential use in orthopedics. While high in-plane-stiffness and strength of these nanomaterials help in toughening, their weaker out-of-plane integrity offers lubrication. Present study investigates effect of aspect ratio of carbon nanotubes (CNT) on toughening and solid-lubrication efficiency of UHMWPE-matrix. A nominal 0.05–0.1wt.% of CNT addition increases hardness and elastic modulus of UHMWPE by 3-45% and 8-42%, respectively. Higher aspect ratio (HAR) CNTs are found more effective in improving hardness and modulus of UHMWPE. Wear rate and friction-coefficient also increase by 530% and 220%, respectively, while reinforced with HAR CNTs. Thermal analysis shows slight increase in crystallinity and stability of composite. HAR CNTs improve interfacial bonding with matrix, due to their morphological similarity to polymer chains, as compared to low aspect ratio CNT. Aspect ratio of CNTs significantly dominates strengthening and tribological behavior of UHMWPE.

Keywords: A. Polymer- matrix composites (PMCs), B. Wear, B. Mechanical properties, D. Thermal analysis.

1. INTRODUCTION

Ultra-High Molecular Weight Polyethylene (UHMWPE) has been extensively used in various joint replacement applications (artificial hip cups and tibial inserts), with articulating counteract surfaces made of metals or ceramics [1, 2]. UHMWPE has been chosen over other commercial polymers because of its superior toughness, impact strength, low friction and abrasive sliding wear resistance. In addition, easy fabrication ability and superior biocompatibility of UHMWPE in the physiological environment makes it even suitable for the above mentioned application [3]. However, tribological behavior of UHMWPE lining plays a critical role in deciding life span of modular acetabular cup in total hip implants [4]. Though UHMWPE is in clinical application for long time, its performance in extended period has been limited due to its inherent low yield strength, leading to permanent deformation, severe wear and ultimately premature failure of implant. Relative motion between the metal/ceramic

femoral head and polyethylene lining surfaces generates wear debris. Generated debris migrates to the tissues around the implant through blood and invites inflammatory reactions, leading to osteolysis and gradual aseptic loosening of the implant [5]. Thus, a drive always exists to improve the stiffness and wear resistance of UHMWPE lining without increasing the coefficient of friction (COF), which is a challenging goal in itself. Over the past decades, significant efforts have been taken up for the enhancement of tribological and mechanical properties of UHMWPE bearing. Various techniques are explored for this purpose, namely, cross linking of polymer [6-8], increasing the degree of crystallinity through ionizing radiation [9] and chemical methods [10]. These modifications decreased the risk of wear rate of UHMWPE. But a negative effect on ductility and strength is noted, resulting into catastrophic failure of the implant [11, 12]. However, main aim with the UHMWPE surfaces to be used in orthopedic joints is to have a unique combination of high wear resistance and low coefficient of friction, which apparently appears contradictory.

Polymer matrix composite is an alternate solution, where UHMWPE matrix is reinforced with either UHMWPE fibers itself [13] or various other carbonaceous nanoparticles and fibers, e.g., nanodiamonds, graphite and carbon nanotubes [3, 14-21]. Different processing routes, namely, solvent casting, ultrasonication, *in situ* polymerization and ball milling, are employed to synthesize these composites with an aim of improving mechanical and wear behavior of UHMWPE matrix [16]. Reinforcing with 5 wt. % of UHMWPE fibers increases the tensile strength (96 MPa) and elastic modulus (1.4 GPa) of the polymer matrix by three times and two times, respectively. Self reinforced UHMWPE composites have shown superior mechanical properties, but wear properties are found to be similar to un-reinforced UHMWPE [13]. Addition of 3 wt% graphene oxide to UHMWPE shows 44% increase in hardness and 70% decrease in wear rate with respect pure UHMWPE [22]. Whereas, addition of 1 wt. % graphene into UHMWPE, has shown 125% increase in elastic modulus and 100% increase in yield stress [14]. On the other hand, a very small amount (0.2 wt. %) of single wall carbon nanotube (SWCNT) addition to UHMWPE increases hardness and elastic modulus by ~ 66% and 58%, respectively [23]. Wear durability is also improved impressively in this composite by more than two orders of magnitude [23]. Hence, among all the reinforcement, carbon nanotube (CNTs) is a promising choice for polymer composite due to their exceptional properties, like, high elastic modulus (200–1000 GPa) and tensile strength (50–200 GPa) [24, 25], which helps in improving the mechanical behavior of polyethylene with minimum addition. As reinforcement to polymer, CNTs can also offer the unique combination of improved toughness and decreased co-efficient of friction on surface by acting as a solid lubricant. However,

effective utilization of CNTs in composite is strongly dependent on their uniform dispersion in the matrix, without disturbing their structural integrity [26]. Homogeneous reinforcement of phases and good interfacial bonding enhance the load transfer efficiency from matrix to CNTs and show improved mechanical properties in the composite [27]. Strong interfacial bonding between polymer matrix and CNT is highly influenced by the aspect ratio of CNTs [28]. Hence, aspect ratio of the reinforcement is considered as a dominating factor for developing the theoretical and experimental models [29]. A few studies have explored the influence of CNT diameter and aspect ratio on mechanical properties of polymer composites. Diameter of CNTs have shown significant effect on fatigue behavior of an epoxy – 2.5 wt.% multi wall carbon nanotube (MWCNT) composite. A 10 fold reduction in the fatigue crack growth rate is observed for CNTs having diameter in the range of 5-8 nm, while only 2 fold improvement was recorded for thicker CNTs with diameter of 20-30 nm [30]. Both the CNTs possessed similar length of 15 – 10 μm . Thus, fatigue resistance and fracture toughness of epoxy is found to be functions of the aspect ratio of the filler phase. Ayatollahi et al. [28] have studied the epoxy-0.5 wt% MWCNT composite with different aspect ratio of CNTs. A maximum of 32% increase in fracture toughness was observed with MWCNTs having aspect ratio of 1000 and it is decreased to 21% for a lower aspect ratio of 455. Both modulus of elasticity and tensile strength were increased with decreasing tube diameter. Small diameter and higher surface to volume ratio of CNTs form a percolated network in the composites and build strong interfacial bond with polymer chains, thus increasing the load transfer sites between the matrix and reinforcement [14].

Tribological properties of CNT reinforced UHMWPE composite is a highly relevant characteristic for potential bearing application. Only a few studies have dealt so far with the influence of CNT concentrations on wear and thermal stability of the polymer composite [31-37]. All of these studies explored effect of CNT addition on the surface hardness, strength, plowing and cutting resistance of UHMWPE matrix. The reports mostly include improvement in wear resistance and increase in coefficient of friction (COF). Few exceptions to this trend were also observed by Wei et al. [36] and Lee et al. [37], who reported decrease in COF by ~20 % with addition of 5 wt % CNTs. This decrease in COF is attributed to the self lubrication offered by CNTs. However, effect of aspect ratio of CNT in controlling the lubrication on polymer matrix surface is not reported.

As mentioned earlier, not many researchers have explored the morphological effect of CNT on mechanical properties of the composite. However, to the best of the authors' knowledge, no studies till date have dealt with the effect of CNT morphology on wear behavior of any polymer based composites. A thorough understanding of the

evolution in wear mechanism in presence of CNTs with different morphology, concentration and distribution is required to comprehend the role played by morphology of reinforcement.

In order to experimentally analyze the effect of CNT morphology on mechanical and tribological behavior of UHMWPE matrix, we used two different aspect ratios of MWCNTs, with two different concentrations, as reinforcement. Ball on disc tribometer and nanoindentation studies were conducted to study necessary properties required for artificial acetabular cup, e.g., COF, specific wear rate, hardness and elastic modulus. Differential scanning calorimetry and thermo gravimetric analysis was also carried out to study the thermal, oxidation or degradation behaviors of the composite structure. Analysis of the results gave us an in-sight on the origin of structural change occurring in composites, due to different aspect ratio of CNTs used as reinforcement. Finally, the obtained results were thoroughly analysed to understand the mechanisms dominating mechanical and tribological behavior of the composite.

2. EXPERIMENTAL PROCEDURE

2.1 Materials and Processing

Medical grade ultra high molecular weight polyethylene (UHMWPE) powder was procured from J.P polymer, Mumbai, India. It was in the form of white powder with an average granular particles size of 120 μm and density of 0.953 g/cm^3 . Two different aspect ratios of MWCNTs with more than 95% purity are used in this study. High aspect ratio CNTs (HARC), with outer diameter of 10-12 nm, length of 8-12 μm and density of 1.8 g/cm^3 were procured from CNano Technology Limited, Beijing, China. Low aspect ratio CNTs (LARC), with outer diameter of 40-70 nm, length of 1-3 μm and density of 2.1 g/cm^3 were procured from Inframat Corporation (Willington, CT). Their average aspect ratios (ratio of average length to diameter) are about 900 and 75, respectively for HARCs and LARCs. Fig. 1a, b and c show FESEM images of as received CNTs as well as UHMWPE powder. An inset in Fig. 1a and b clearly reveals the difference in aspect ratios of two types of CNTs used in this study. Initially, CNTs were homogeneously dispersed in acetone using a probe sonicator for 60 minutes (power - 500 Watts; frequency - 20 kHz) followed by another 30 minutes sonication after mixing the UHMWPE powder to dispersed CNT solution. In order to avoid the agglomeration of CNT, polymer powder was added in dispersed CNTs slowly, while keeping ultrasonication on. In this process, CNTs, due to their high surface energy, gets attached to the polymer powder surface and while settling down they get uniformly distributed. The composite powders were dried completely in